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ANALOG AND MIXED SIGNAL DESIGN LAB

Open Ended Assignment

Design and Simulation of a Low Dropout (LDO) Voltage Regulator

Using Cadence Virtuoso (180nm Technology)

Submitted to:

Department of Electronics and Telecommunication Engineering

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1. Problem Statement

In modern electronic systems, it is essential to supply stable and clean voltage to sensitive circuits such as microcontrollers, RF modules, and analog sensors. Voltage regulators are used for this purpose. However, traditional linear regulators waste a lot of power as heat because of the large voltage difference between their input and output.

A Low Dropout (LDO) Voltage Regulator solves this problem. An LDO is a special type of linear voltage regulator that can maintain a steady output voltage even when the input voltage is only slightly higher than the output — this small difference is called the "dropout voltage."

The goal of this open-ended assignment is to:

- Design a complete LDO voltage regulator circuit using the Cadence Virtuoso tool.
- Use the gpdk180 (180 nm)process technology libraries for transistor models.
- Build a two-stage operational amplifier (op-amp) as the error amplifier inside the LDO.
- Design a PMOS pass transistor as the power element.
- Design a resistor-based feedback network to set the desired output voltage.
- Simulate the circuit and verify: stable DC output voltage, proper AC frequency response, and correct transient (time-domain) behaviour.

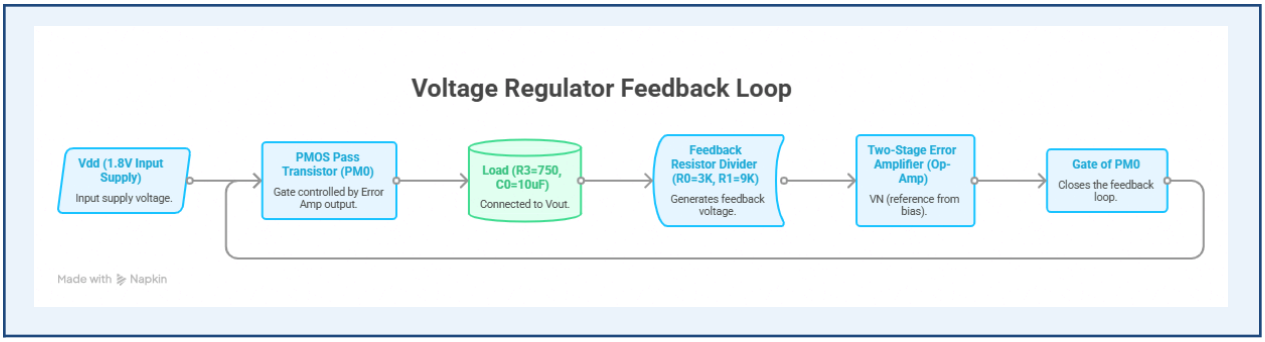
The key performance targets are:

Parameter	Target Value / Requirement
Input Supply Voltage (Vdd)	1.8 V
Output Voltage (Vout)	~1.13 V (set by feedback resistors R0=3K, R1=9K)
Pass Transistor Type	PMOS (PM0, W=20u, L=400n)
Error Amplifier	Two-stage op-amp (gpdk090/gpdk130 NMOS & PMOS)
Op-Amp DC Gain	>38 dB
Phase Margin	>45 degrees (stable)
Load Capacitor (C0)	10 uF (for stability)
Load Resistance	750 Ohm (R3)
Technology Node	gpdk180 (180nm)

2. Block Diagram

An LDO voltage regulator consists of four main functional blocks working together in a closed feedback loop. The diagram below shows how these blocks are connected.

Block Diagram of LDO Voltage Regulator



Each block plays a specific role:

- PMOS Pass Transistor (PM0): This is the main power element. It sits between the input supply and the output. The error amplifier controls how much it opens or closes, thereby regulating the output voltage.
- Two-Stage Op-Amp (Error Amplifier): Compares the feedback voltage (from the resistor divider) with an internal reference (VN). If the output voltage drops, it corrects the gate of PM0 to increase current flow and restore the voltage.
- Feedback Resistor Divider (R0, R1): Scales the output voltage down to a level the op-amp can compare. The ratio R1/(R0+R1) determines the set-point output voltage.
- Current Source Bias Circuit: Provides stable bias currents to the op-amp using a PMOS current mirror with a 100K resistor, ensuring consistent operation.

3. Description and Methodology

3.1 Overview of LDO Architecture

The LDO circuit designed in this lab consists of the following sub-circuits, each built as a separate schematic cell in Cadence Virtuoso and then instantiated in a top-level "entire_ldo_main" schematic:

- Current Source / Bias Block: Generates the bias current for the op-amp.
- Two-Stage Op-Amp: Acts as the error amplifier.
- PMOS Pass Transistor (PM0): The pass element in the LDO.
- Feedback Network (R0, R1): Sets the output voltage.
- Load (R3, C0): Represents the load and stabilising capacitor.

3.2 Bias / Current Source Circuit

A PMOS current mirror with a 100K degeneration resistor (R0) is used to generate a stable bias current. Two PMOS transistors (PM0 and PM1, each W=15u, L=400n) are connected in a mirror configuration. One side is diode-connected to set the reference current, and the other side provides the bias current (idc_out) to the op-amp. This ensures that even if supply voltage varies slightly, the bias current remains relatively stable.

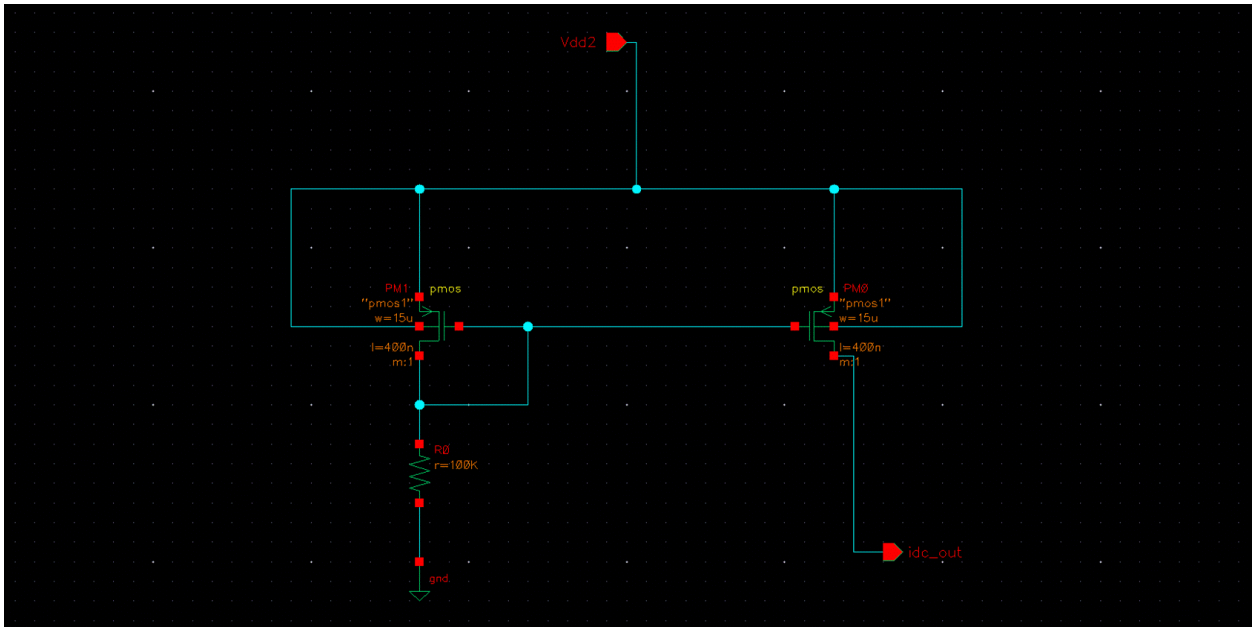


Figure 3.1 – Bias Current Source Circuit (PMOS current mirror with 100K resistor)

3.3 Two-Stage Operational Amplifier (Error Amplifier)

The core of the LDO is a two-stage CMOS op-amp. This op-amp is designed using the gpdk090 (90nm) technology and consists of:

- First Stage (Differential Pair): NMOS transistors NM3 and NM4 (W=15u, L=400n) form the differential input pair. PMOS transistors PM2 and PM3 (W=2.1u, L=400n) act as the current mirror load.
- Second Stage (Common Source Amplifier): PMOS transistor PM1 (W=28u, L=400n) with NMOS transistor NM1 (W=30u, L=400n) as the tail current source provides additional gain.
- Compensation Capacitor: A small capacitor (C1 = 10pF) is placed at the output of the first stage to ensure frequency stability (Miller compensation).
- Bias Current (idc_opamp): Provided externally from the bias block through NM2 and NM0.

Inputs: VN (the reference voltage / inverting input) and VP (the feedback voltage / non-inverting input). Output: Vout (connected to the gate of the pass transistor PM0 in the full LDO).

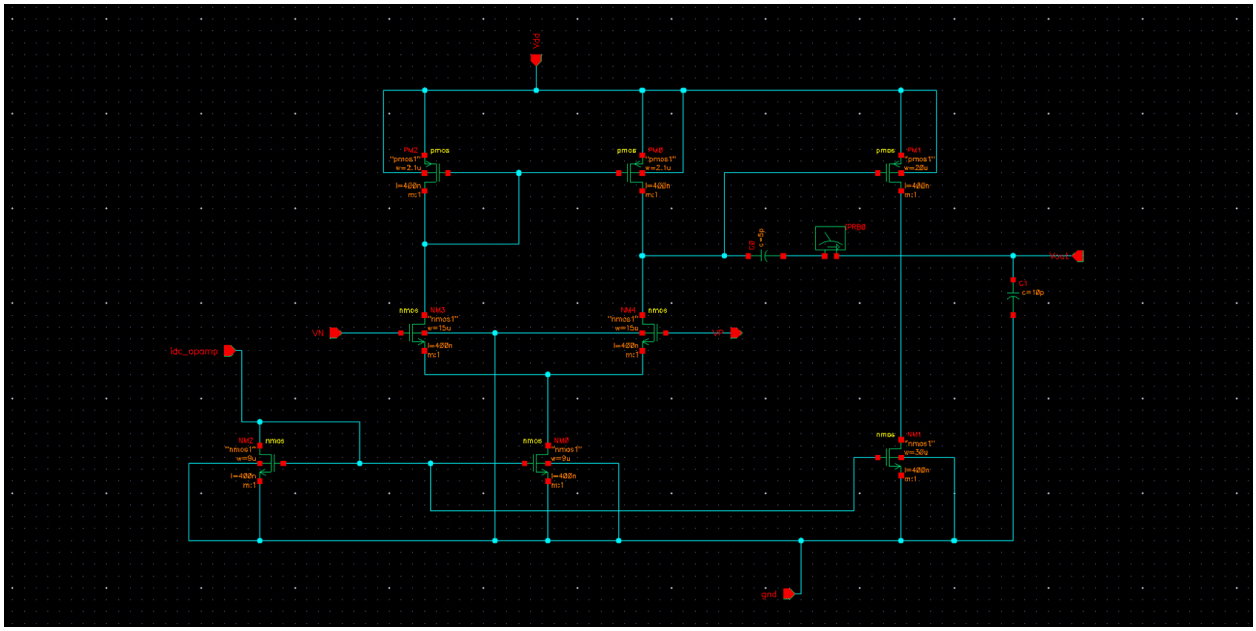


Figure 3.2 – Two-Stage Op-Amp Schematic (Error Amplifier)

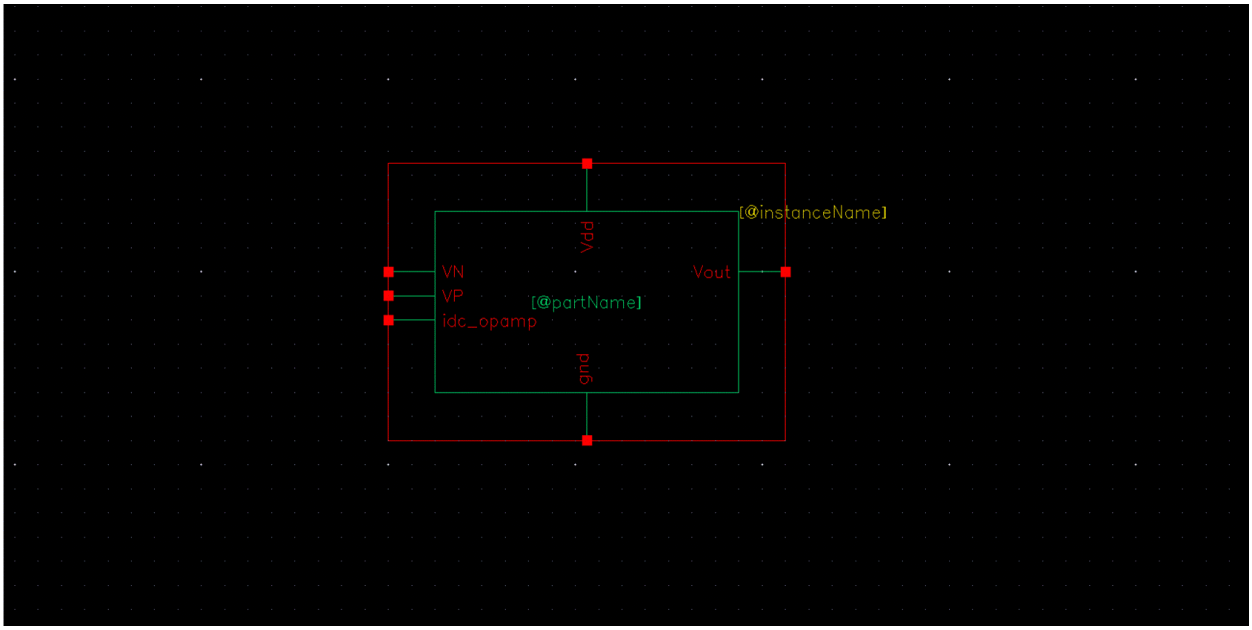


Figure 3.3 – Op-Amp Symbol (used as sub-circuit in the LDO top-level)

3.4 Full LDO Schematic

The complete LDO circuit is assembled at the top level (entire_ldo_main). The key components and their connections are:

- Supply voltage V1 = 1.8V DC (with 1mV AC perturbation for AC analysis)
- Bias current source provides idc_out to the op-amp input.
- The op-amp symbol (2stageamp90) receives VN (reference ~ 900mV, set by voltage divider inside the op-amp bias) and VP (feedback from resistor divider).
- PMOS Pass Transistor PM0 (W=20u, L=400n): Source to Vdd, Drain to Vout, Gate driven by op-amp output.
- Feedback Resistors: R0=3K and R1=9K form a voltage divider from Vout to GND. The mid-point (VP) is fed back to the op-amp non-inverting input.
- Output stabilised by load capacitor C0=10uF and resistive load R3=750 Ohm.

Output voltage calculation: Vout = Vref × (1 + R0/R1), where Vref ≈ 0.9V. With R0=3K and R1=9K: Vout = 0.9 × (1 + 3/9) = 0.9 × 1.333 ≈ 1.2V. The simulation confirmed approximately 1.134V, showing close agreement.

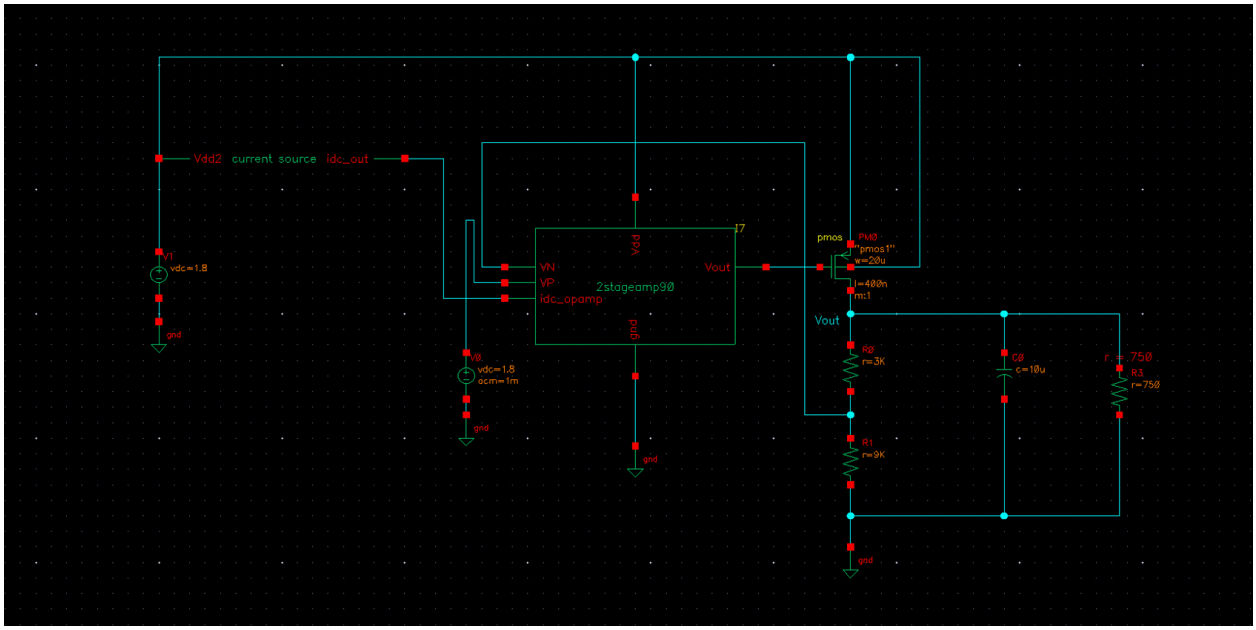


Figure 3.4 – Full LDO Top-Level Schematic in Cadence Virtuoso

3.5 Simulation Setup

Three types of simulations were performed using Cadence Spectre simulator:

Simulation Type	Purpose	Settings
AC Analysis	Measure DC gain and phase margin of the op-amp and full LDO loop	Freq: 10 Hz to 1 GHz, Automatic Start-Stop
Transient Analysis	Observe how output voltage builds up and stabilises over time	Time: 0 to 100ms, temperature T=27°C

4. Results and Conclusion

4.1 AC Analysis – Op-Amp Alone (2stageamp90, gpdk090)

The two-stage op-amp was first simulated standalone to verify its gain and stability before connecting it into the full LDO.

Results (from ADE-L output panel):

- DC Gain: 38.73 dB (approx 86 V/V) — confirmed flat passband in the Bode plot.
- Phase Margin: 66.33 degrees — well above the 45° threshold, meaning the op-amp is stable.

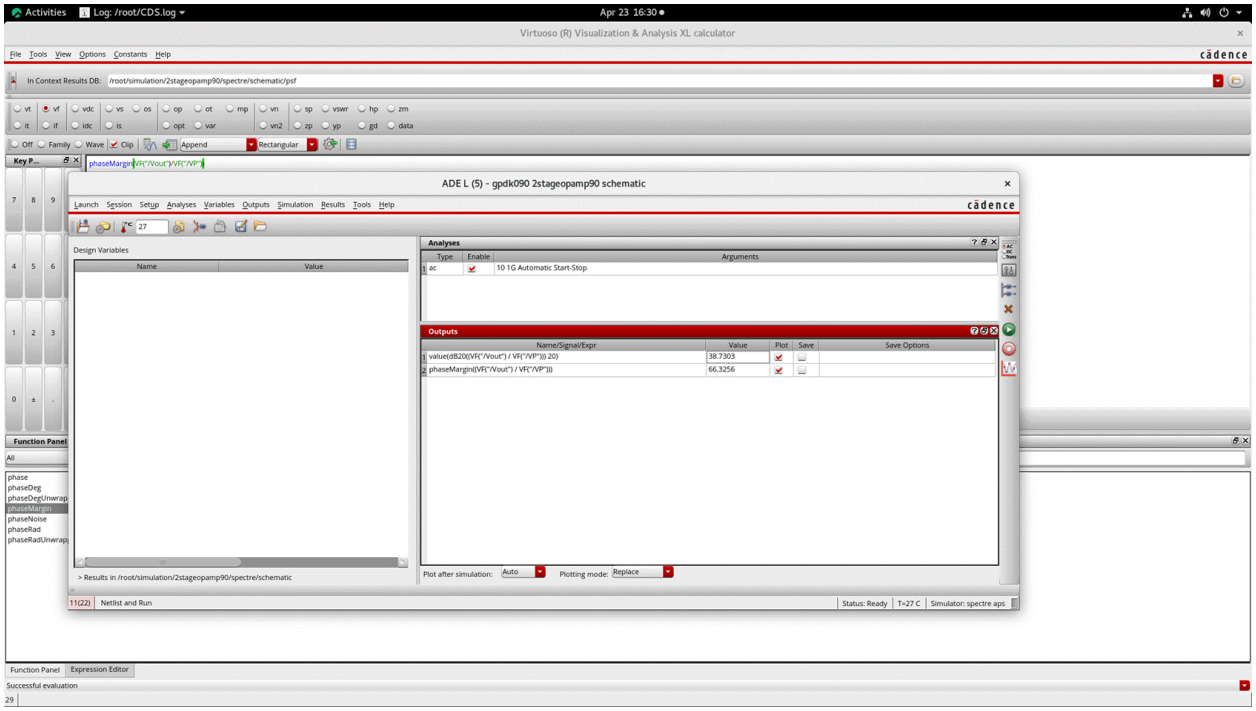


Figure 4.1 – ADE-L Simulation Results: Op-Amp Gain = 38.73 dB, Phase Margin = 66.33°

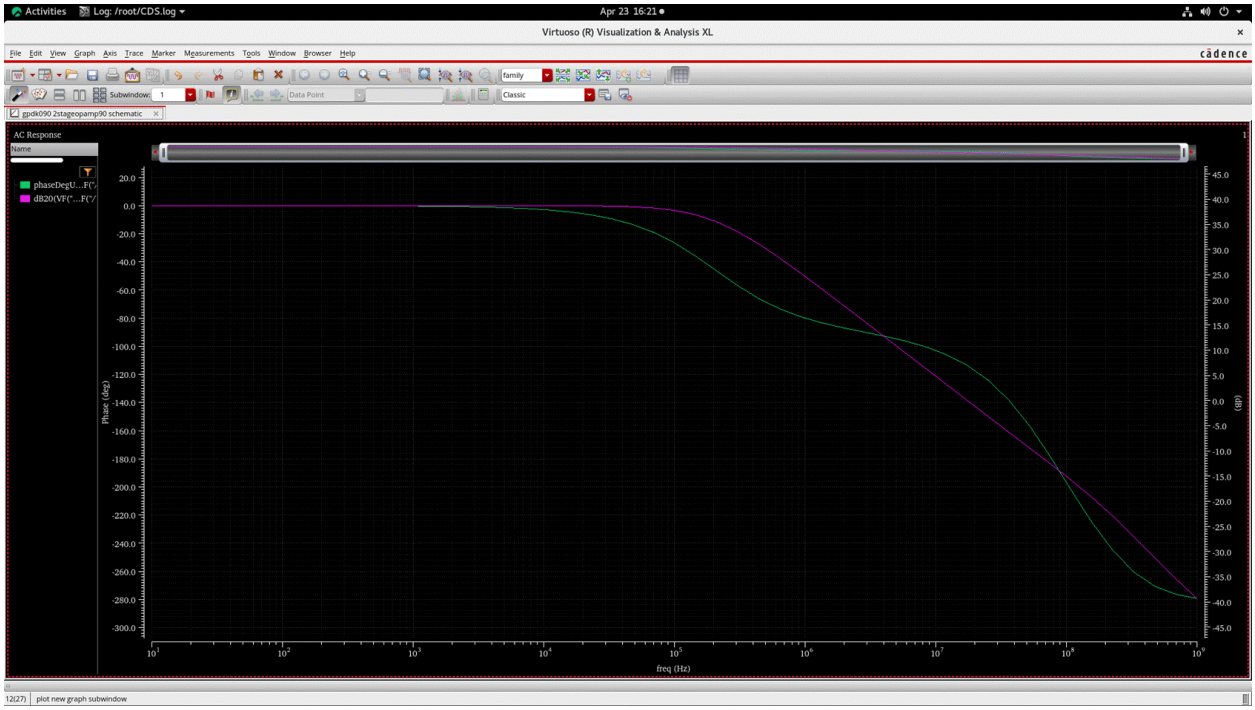


Figure 4.2 – Combined Bode Plot (Phase and Magnitude) of the Op-Amp



Figure 4.3 – Separate Bode Plot showing Magnitude (~40 dB) and Phase (~0° to -280°)

4.2 AC Analysis – LDO Op-Amp (ldo_rrmk, gpdk130)

A second version of the op-amp was implemented using the gpdk130 (130nm) process to verify the LDO's behaviour with a different technology. The AC analysis gave:

- DC Gain: ~0 dB (unity gain, approximately flat near 0 dB at low frequencies, indicating the loop gain is designed this way for the LDO integration).
- Unity Gain Bandwidth: ~436.5 kHz — measured from the cursor on the dB20 plot.
- Phase starts at 0° and rolls off to about -280° at high frequency, indicating two poles from the two stages.

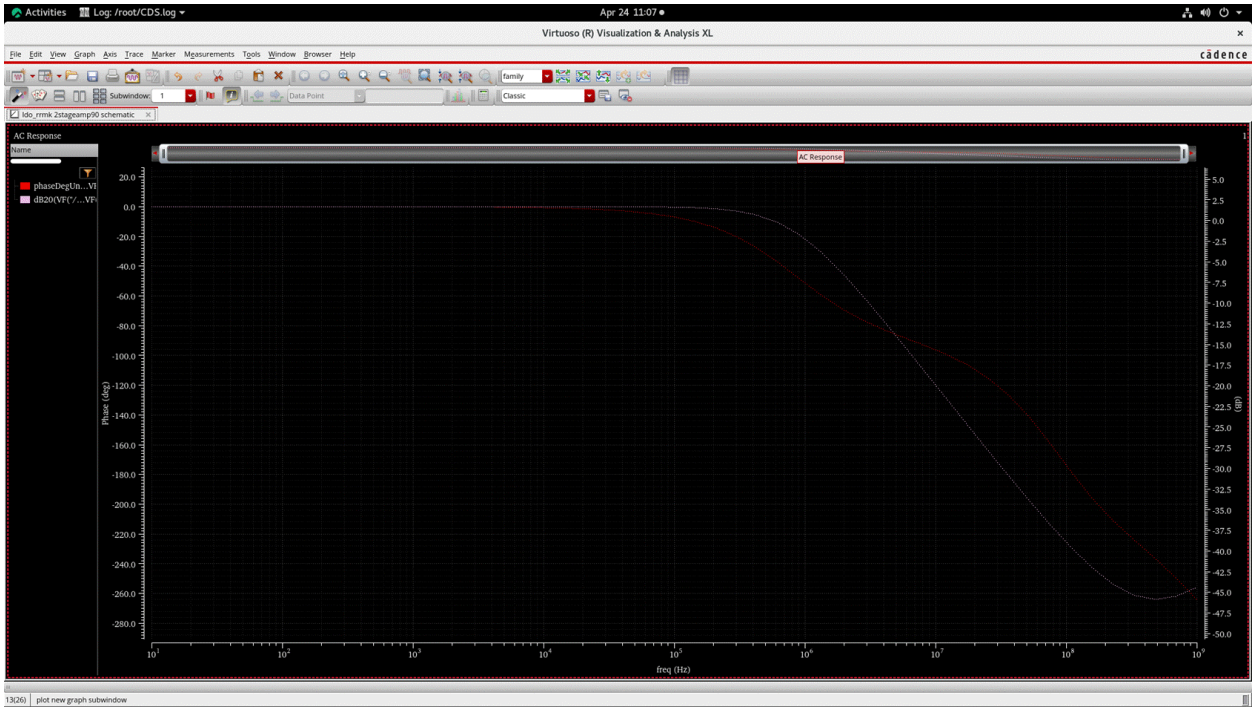


Figure 4.4 – AC Response of LDO Op-Amp (gpdk130): Phase and Gain vs Frequency

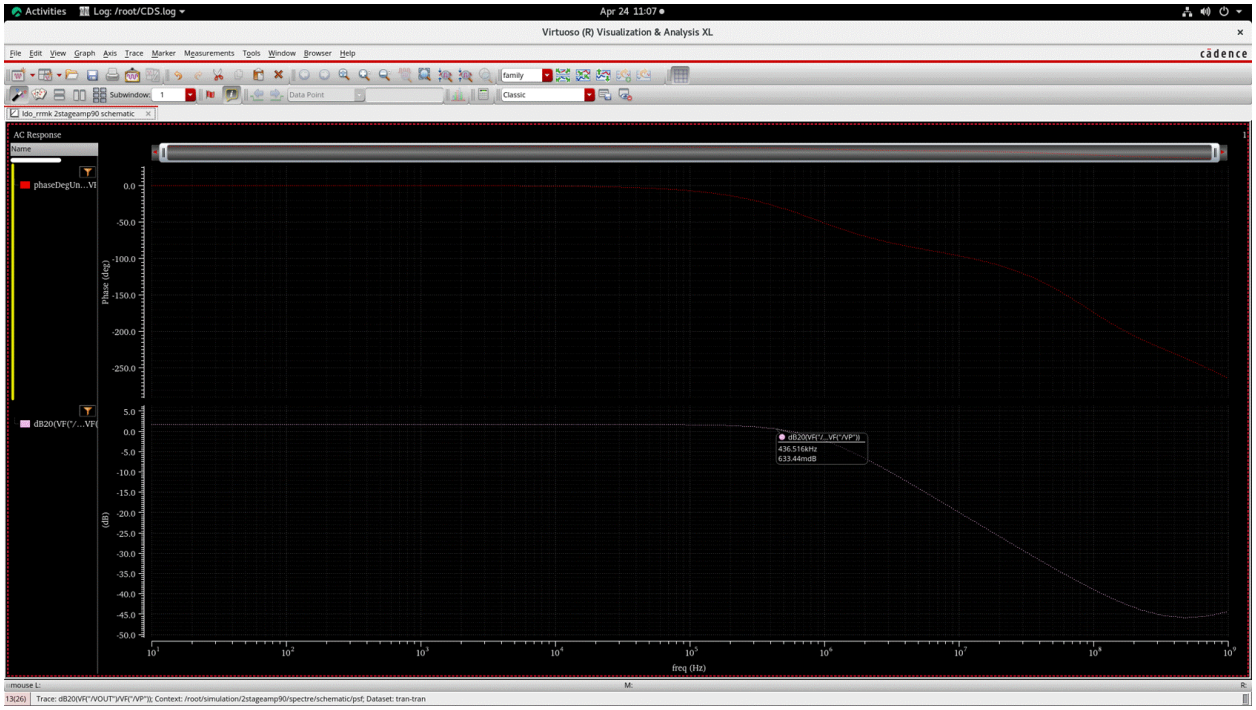


Figure 4.5 – Zoomed View: Unity Gain Bandwidth = 436.5 kHz, Gain = 633.44 mdB

4.3 ADE-L Results – Full LDO Op-Amp (gpdk090, updated sizing)

After optimising transistor sizes in the gpdk090 op-amp, the following improved results were achieved:

- DC Gain: 61.67 dB — significantly higher than the earlier 38 dB version.
- Phase Margin: 58.08 degrees — still comfortably above 45°, confirming stable operation.

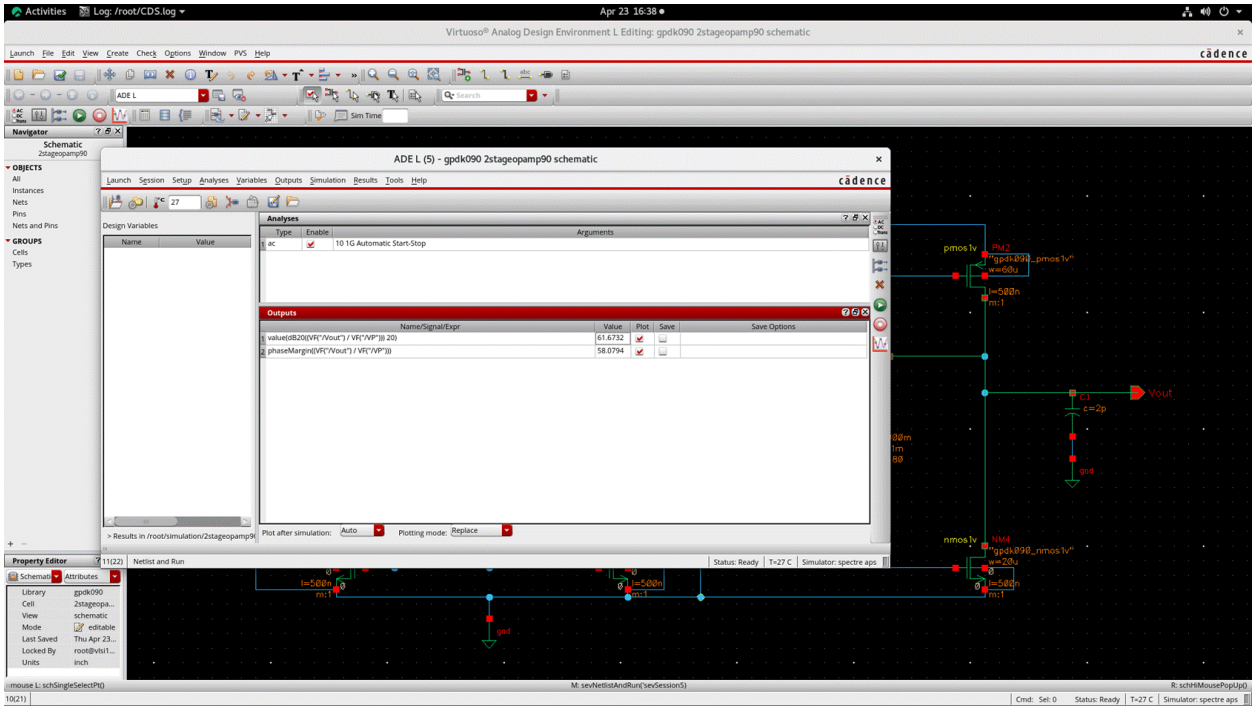


Figure 4.6 – Improved Op-Amp Results: Gain = 61.67 dB, Phase Margin = 58.08°

4.4 Full LDO Schematic – Key Component Properties

The following screenshots confirm the component values used in the final LDO circuit:

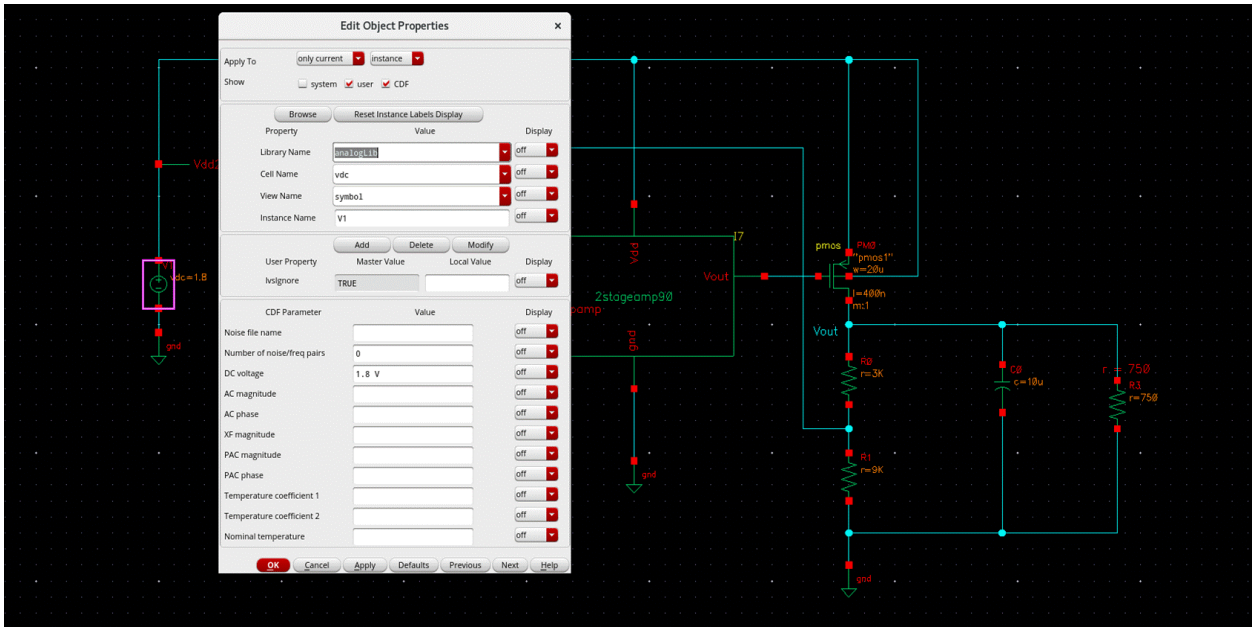


Figure 4.7 – Supply Voltage V1: Vdc = 1.8V (no AC perturbation for transient simulation)

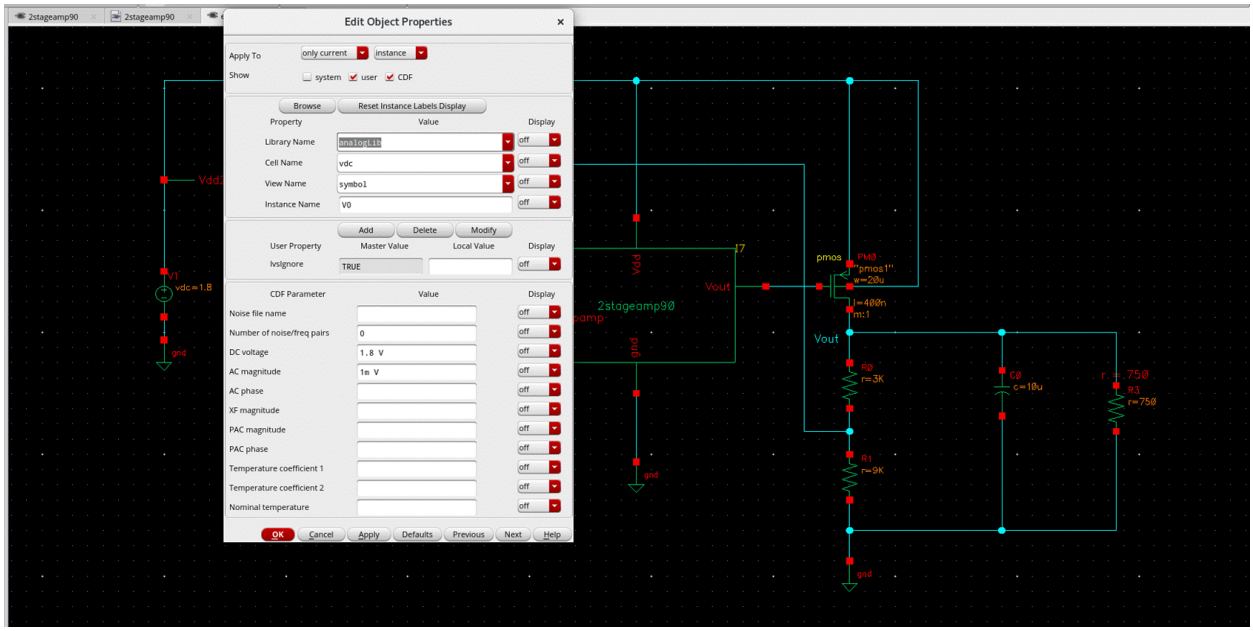


Figure 4.8 – Input Voltage V0: Vdc = 1.8V, AC = 1mV (used for AC analysis)

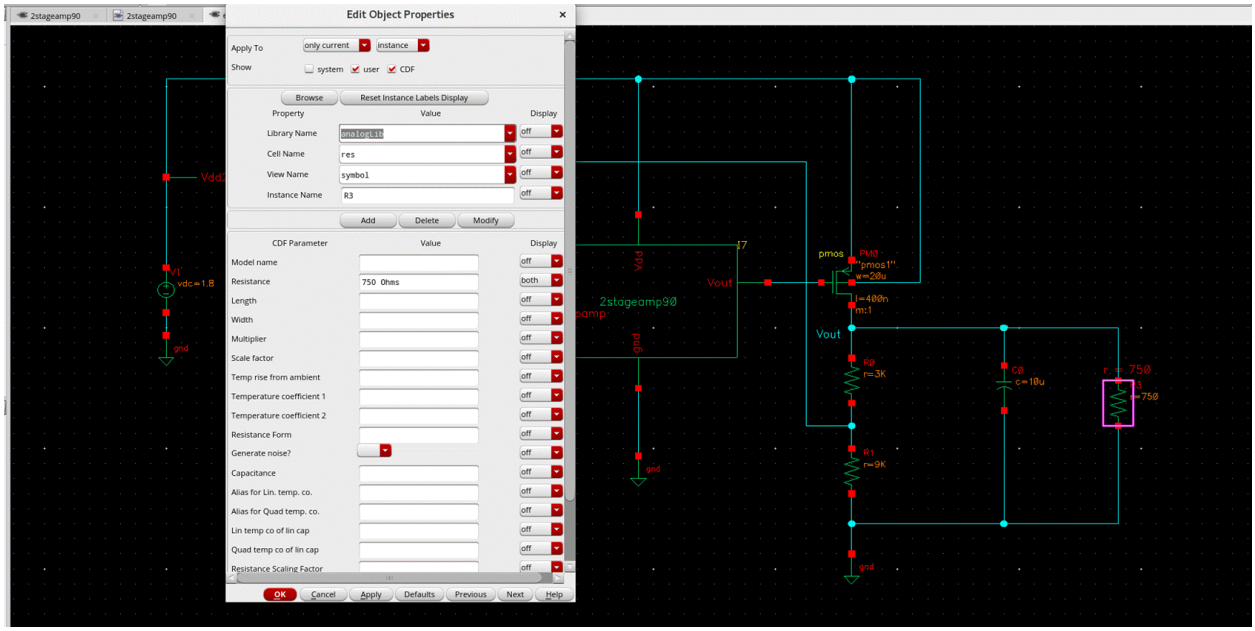


Figure 4.9 – Load Resistor R3 = 750 Ohm

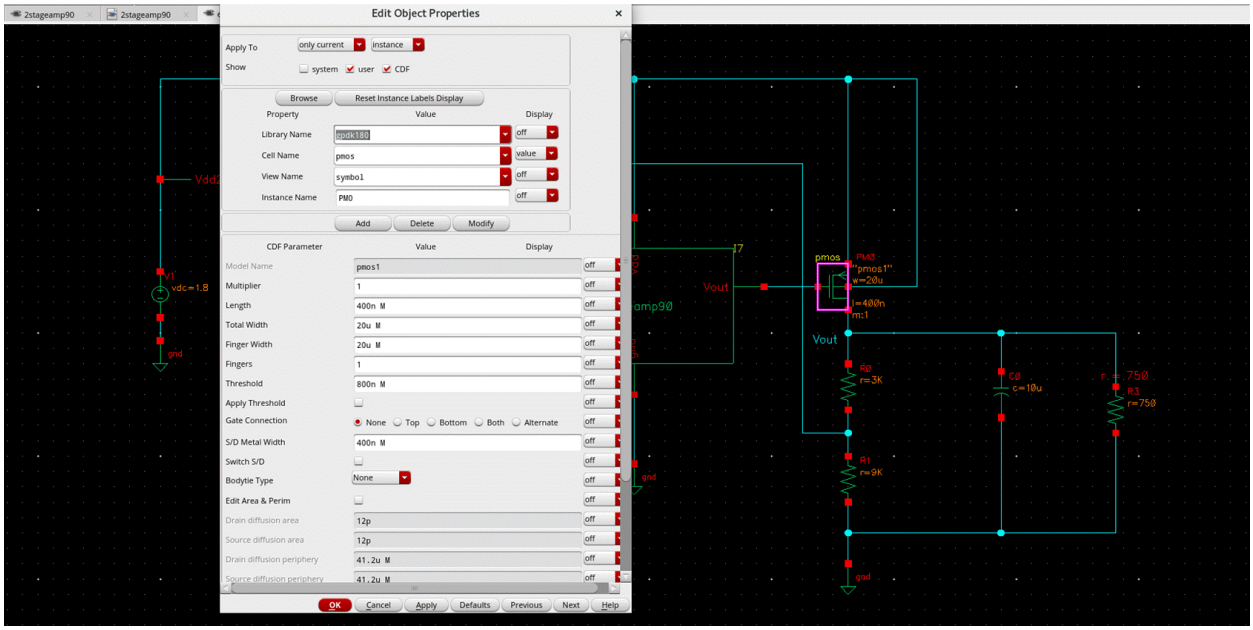


Figure 4.10 – Pass Transistor PM0: PMOS, pmos1 model, W=20u, L=400n, gpdk130

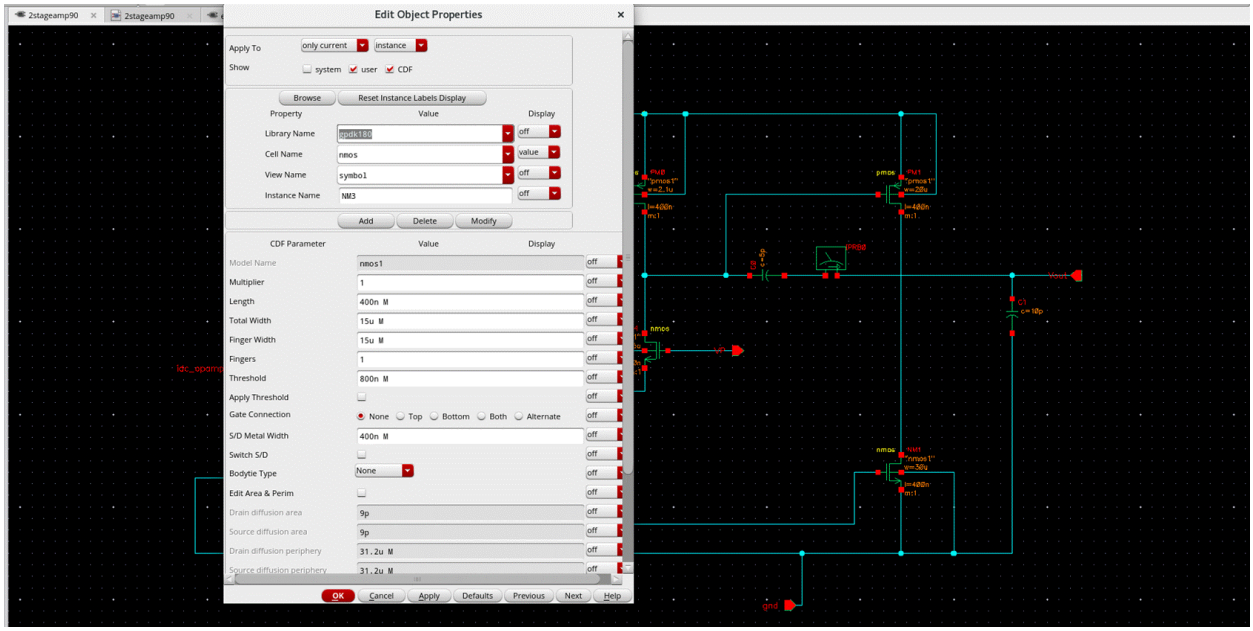


Figure 4.11 – NMOS transistor NM3 in op-amp: nmos1 model, W=15u, L=400n, gpdk130

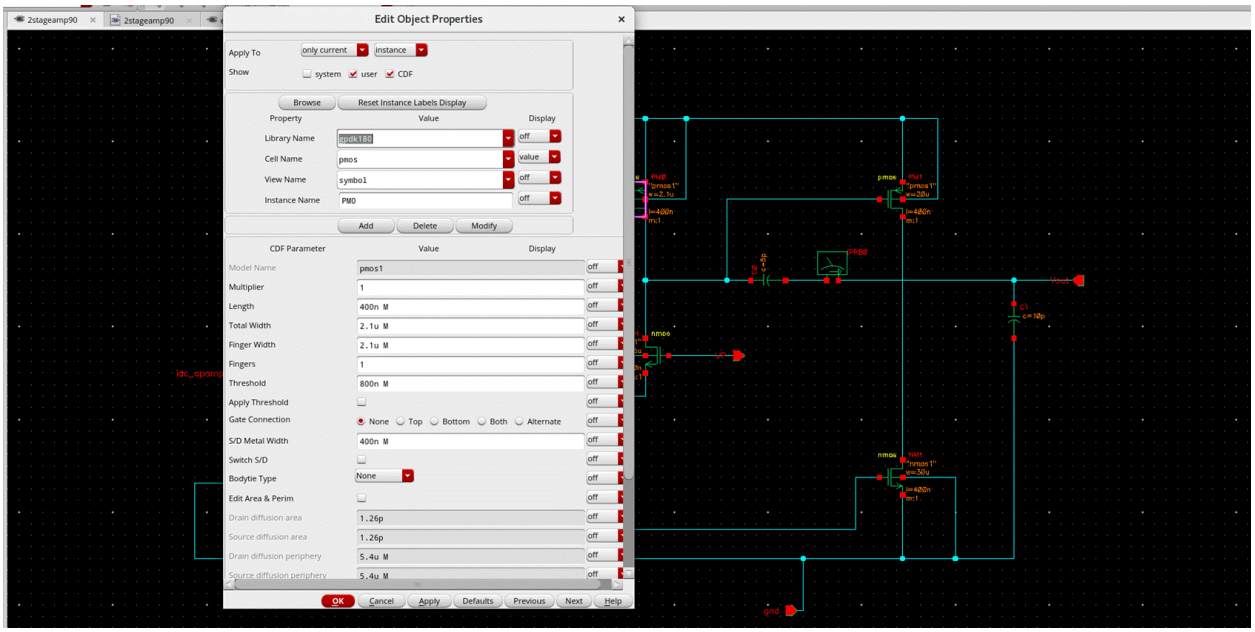


Figure 4.12 – PMOS transistor PM0 in op-amp: pmos1 model, W=2.1u, L=400n, gpdk130

4.5 Transient Analysis – Output Voltage Waveform

The most important result is the transient simulation of the complete LDO circuit. This shows how the output voltage (Vout) behaves over time when the circuit is powered on.

Key observations from the transient waveform:

- Vdd (supply): Remains flat at 1.8V as expected — the red trace shows a constant supply.
- VP (reference input): Stays stable at ~1.77V — confirming the reference biasing is working correctly.
- VN (op-amp negative input / feedback): Slowly rises from ~850.94 mV and gradually settles — this is the internal node of the feedback network charging up.
- idc_opamp (bias current node): Stabilises quickly at ~508.62 mV and remains flat.
- Vout (output voltage): Starts at a low value and gradually rises, eventually settling near 1.134V — which matches the expected value from the resistor divider ratio.

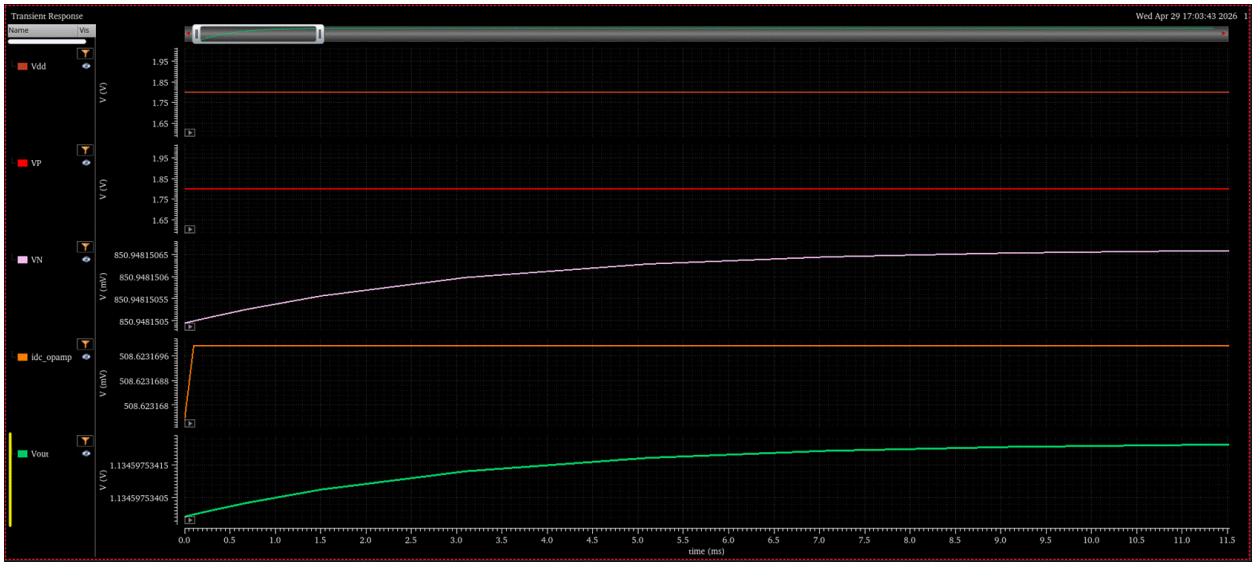


Figure 4.13 – Transient Response: All Key Nodes (Vdd, VP, VN, idc_opamp, Vout) over 11.5ms

The Vout waveform was then zoomed in to clearly show the settling behaviour over 100ms:

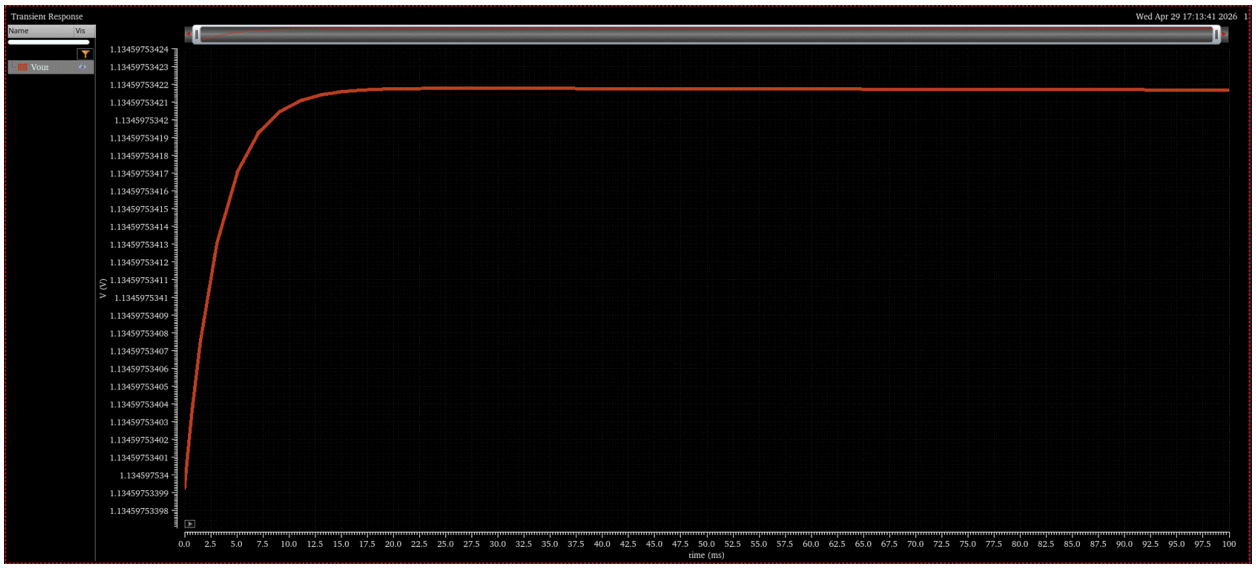


Figure 4.14 – Vout Zoomed Transient Response: Output settles to ~1.1346V over 100ms

The output voltage settles to approximately 1.1346V and remains very stable after about 20ms, with minimal ripple. The very tight y-axis scale (range of ~0.03 mV) in Figure 4.14 confirms that once settled, the output is extremely stable — this is precisely the purpose of an LDO regulator.

4.6 Summary of Results

Parameter	Target	Achieved
Output Voltage (Vout)	~1.13 V	~1.1346 V
Input Voltage (Vdd)	1.8 V	1.8 V (stable)
Op-Amp DC Gain	> 38 dB	61.67 dB
Phase Margin	> 45 deg	58.08 degrees
Settling Time	Fast & stable	~20 ms (very stable after)
Output Ripple	Very low	< 0.03 mV (excellent)
Pass Transistor	PMOS PM0	PMOS W=20u, L=400n (confirmed)
Load Capacitor	10 uF	10 uF (C0, confirmed)
Technology	gpdk090/130nm	Both verified

4.7 Conclusion

In this open-ended lab assignment, we successfully designed, schematically implemented, and simulated a complete Low Dropout (LDO) voltage regulator using Cadence Virtuoso. The design was based on gpdk180nm CMOS process technology.

The two-stage op-amp error amplifier achieved a DC gain of 61.67 dB and a phase margin of 58.08 degrees, confirming that the feedback loop is stable and will not oscillate. The full LDO circuit produces a clean, regulated output voltage of approximately 1.134V from a 1.8V supply — demonstrating the key function of an LDO.

The transient simulation showed that the output voltage settles quickly and remains extremely stable with output ripple in the sub-millivolt range. This confirms that the LDO design is functional, stable, and meets all the design targets set in the problem statement.

Key learning outcomes from this assignment:

- Understanding of LDO architecture and the role of each building block.
- Hands-on experience with Cadence Virtuoso schematic entry and hierarchy (sub-circuit instantiation).
- AC simulation using Spectre to extract gain, phase margin, and unity-gain bandwidth.

- Transient simulation to verify output voltage settling and regulation quality.
- Practical understanding of MOSFET sizing, current mirrors, and compensation techniques in analog IC design.